Date: _____

Ghazi University
(Dera Ghazi Khan)

Assignment

Topic: Programming Lab 2

Roll No: 01

Section: BSCS-C

Submitted To: Sir Asad

Submitted by: Sana Ishaq.

www.btpublishers.com

CamScanner

Date:

Q:1:

write a program that takes an integer and from the user and checks whether the number is "positive", "Negative" or "Zero"?

Logic:

if number > 0 -> positive, if number < 0 -> Negative, else zero.

Input:

```
#include <iostream>
using namespace std;
int main() {
int num;
cout << "Enter a numbe: ";
    cin >> num;
if (num > 0) {
      cout << "positive";
    }
  else if(num < 0) {
      cout << "Negative";
      } else {
        cout << "zero";
      }
    return 0;
  }
```

www.btpublishers.com

Date: _____

Output:

Enter a number: 5;
positive
Enter a number: -3
Negative.
Enter a numbe 0
zero.

CamScanner

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter an integer: ";
    cin >> num;

    if (num % 2 == 0)
        cout << "Even";
    else
        cout << "Odd";

    return 0;
}
```

Tab | { | } | : | ; | "

CamScanner



CamScanner

Date: _____

Q:2: write a program that takes an integer from the user and check whether the number is even or odd?

Logic: Agar number ko 2 pr devide krny pe remainder 0 aye to Even, warna odd.

Input:

```
# include <iostream>
using namespace std;
int main ( ) {
    int num;
cout << "Enter an integer: " ;
    cin >> num;
if (num % 2 == 0)
        cout << "Even";
  else
        cout << "odd";
    return 0;
        }
```

Output:

Enter an integer : 8 positive. (Even)

Enter an integer : 5 Negative (Odd)

CamScanner

new*

```cpp
#include <iostream>
using namespace std;

int main() {
    int marks;

    cout << "Enter marks: ";
    cin >> marks;

    if (marks >= 50)
        cout << "Pass";
    else
        cout << "Fail";

    return 0;
}
```

Tab | { | } | : | ; | "

CamScanner



CamScanner

Date: _____

Q: 3

write a program that takes a student marks as input. if the marks are "50 or above" display pass ; otherwise display fail.

Logic:

if marks >= 50 -> pass.

Input:

```
# include <iostream>
using namespace std;
int main ( ) {
int marks ;
cout << "Enter marks : " ;
cin <> marks ;
```


CamScanner

Date: _____

```
if (marks >= 50) {
        cout << "pass";
} els {
        cout << "fail";
}
    return 0;
}
```

Output:

Enter marks : 60

Pass.

```cpp
#include <iostream>
using namespace std;

int main() {
    int num1, num2;

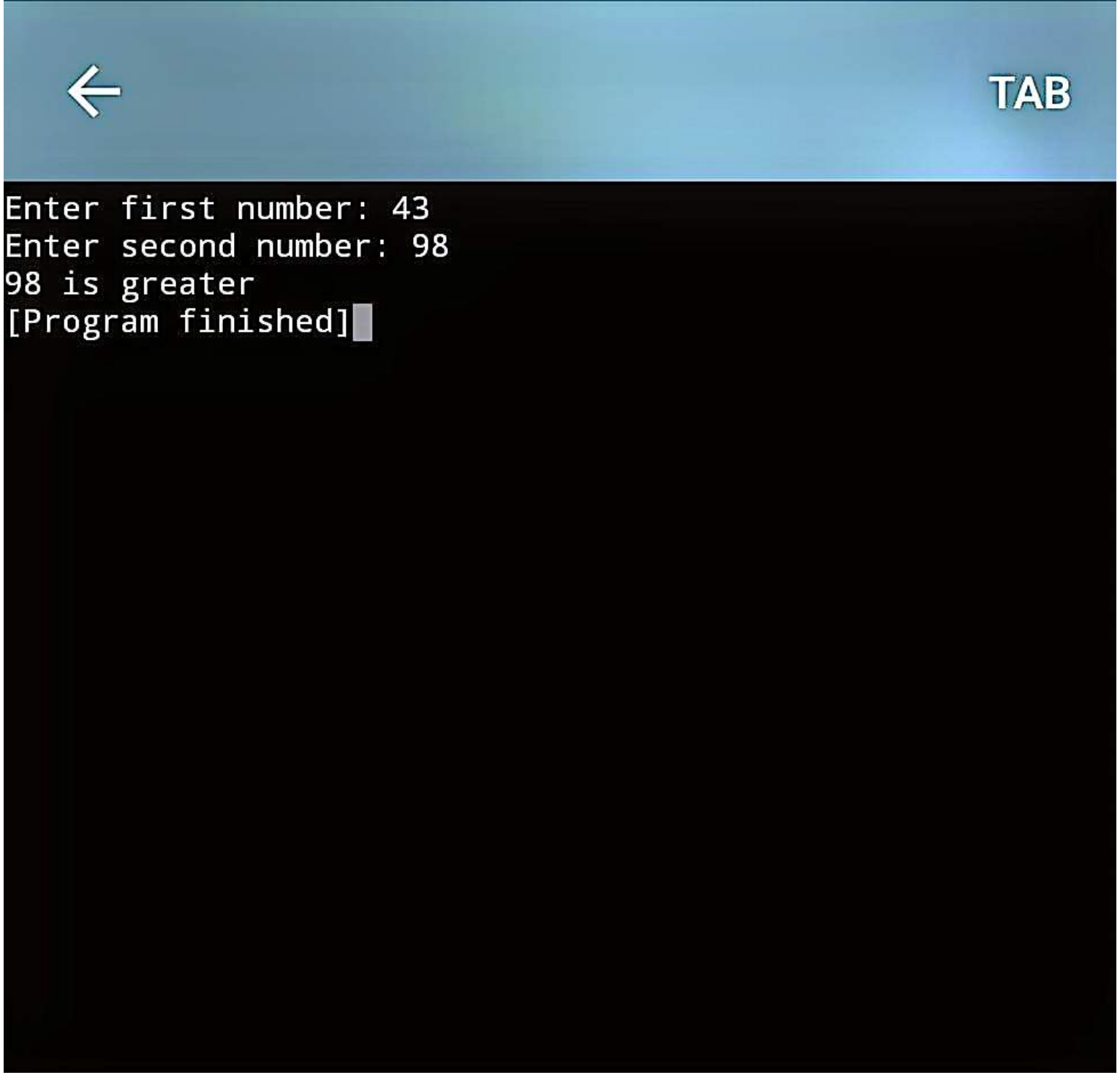
    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    if (num1 > num2)
        cout << num1 << " is greater";
    else if (num2 > num1)
        cout << num2 << " is greater";
    else
        cout << "Both numbers are equal";

    return 0;
}
```

CamScanner


TAB
Enter first number: 43
Enter second number: 98
98 is greater
[Program finished]

CamScanner

Q 4:

write a program that takes *two integers* from d user and displays which number is greater. if both numbers are equal, display both numbers are equal.

Logic:

compare two numbers.

Input:

```
#include <iostream>
using namespace std;
```

Date: _____

```
int main ( ) {
    int a, b;
cout << "Enter two numbers:";
    cin >> a >> b;
if (a > b) {
cout << a << " is greater";
    }
else if (b > a) {
    cout << b << " is greater";
        }
else {
        cout << "Both numbers are equal";
                }
return 0;
    }
```

Output:

```
Enter first number : 43
Enter second number : 98

98 is greater.
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int marks;

    cout << "Enter marks: ";
    cin >> marks;

    if (marks >= 80 && marks <= 100)
        cout << "Grade A";
    else if (marks >= 70)
        cout << "Grade B";
    else if (marks >= 60)
        cout << "Grade C";
    else if (marks >= 50)
        cout << "Grade D";
    else
        cout << "Grade F";

    return 0;
}
```

CamScanner



CamScanner

Date: _____

Q 5:

write a program that takes a student's marks and displays the grade according to the following criteria;

- 80 - 100 → A
- 70 - 79 → B
- 60 - 69 → C
- 50 - 59 → D
- Below 50 → F

Logic:

check from highest grade to lowest.

Input:

```
#include<iostream>
using namespace std;
int main()
{
int marks;
cout << "Enter marks:";
cin >> marks;
if (marks >>= 80 && marks <= 100) {
cout << "Grade: A";
```


CamScanner

Date: _____

```
    }
else if (marks >= 70) {
        cout << "Grade B";
    }
else if (marks >= 60) {
        cout << "Grade C";
    }
else if (marks >= 50) {
        cout << "Grade : D";
    }
else if {
        cout << "Grade:F";
    }
return 0;
}
```

Output:

```
Enter marks : 65
Grade C
```

www.btpublishers.com

CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num;
6      cout << "Enter a five-digit number: ";
7      cin >> num;
8
9      int d1 = num / 10000;
10     int d2 = (num / 1000) % 10;
11     int d3 = (num / 100) % 10;
12     int d4 = (num / 10) % 10;
13     int d5 = num % 10;
14
15     cout << d1 << "   " << d2 << "   " << d3 << "   "
       << d4 << "   " << d5;
16
17     return 0;
18 }
```

CamScanner



CamScanner

Date: _____

Bonus task:

Bonus task : Break number.

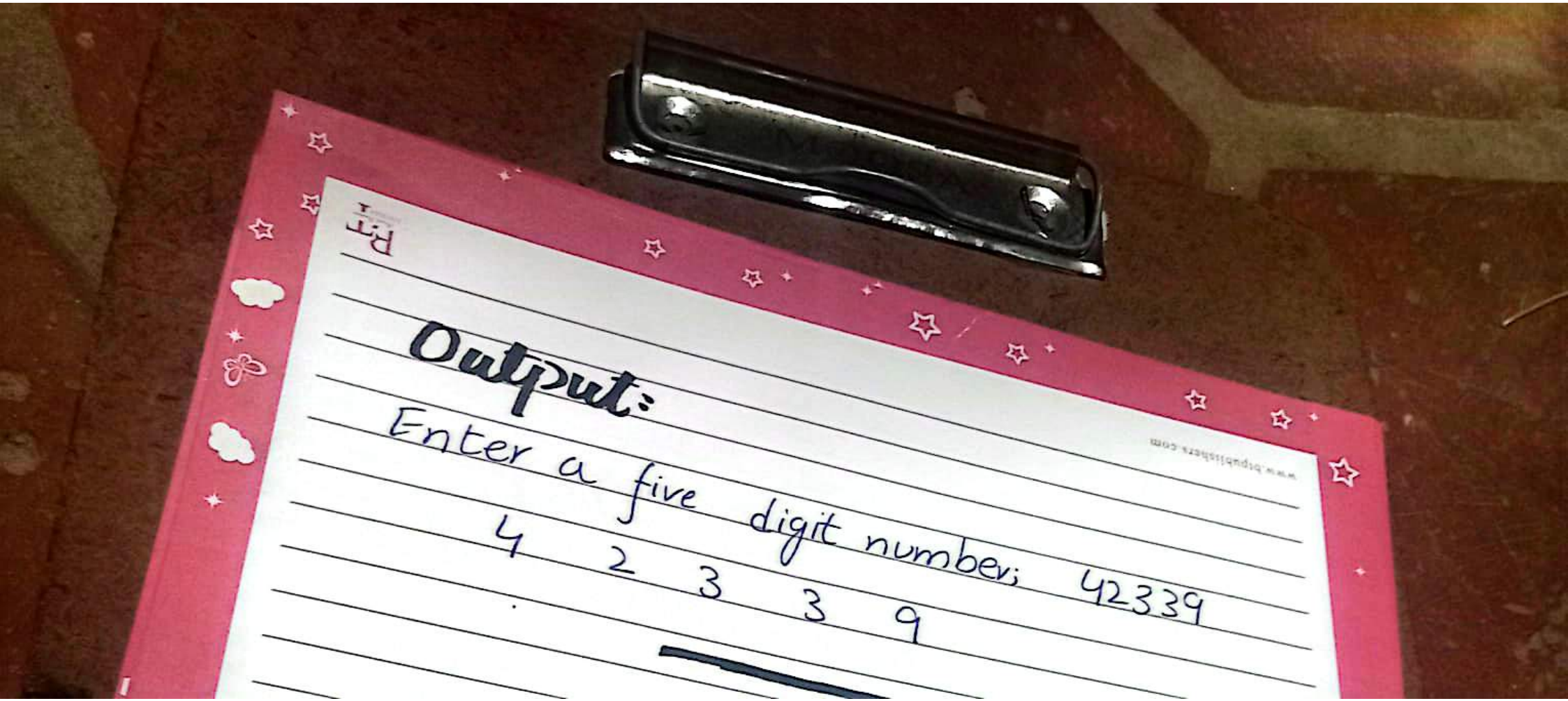
Logic:

/ se pehla digit nikalte hain,
% se baqi digit.

Input:

```
# include <iostream>
Using namespace std;
int main ( )
{
int num;
cout << "Enter a five digit number: ";
    cin >> num;
int d1 = num / 10000 ;
int d2 = (num / 1000) % 10;
int d3 = (num / 100) % 10;
int d4 = (num / 10) % 10;
int d5 = num % 10;
cout << d1 << " " << d2 << " " <<
d3 << " " << d4 << " " << d5;
    return 0;
}
```

www.bfpublishers.com

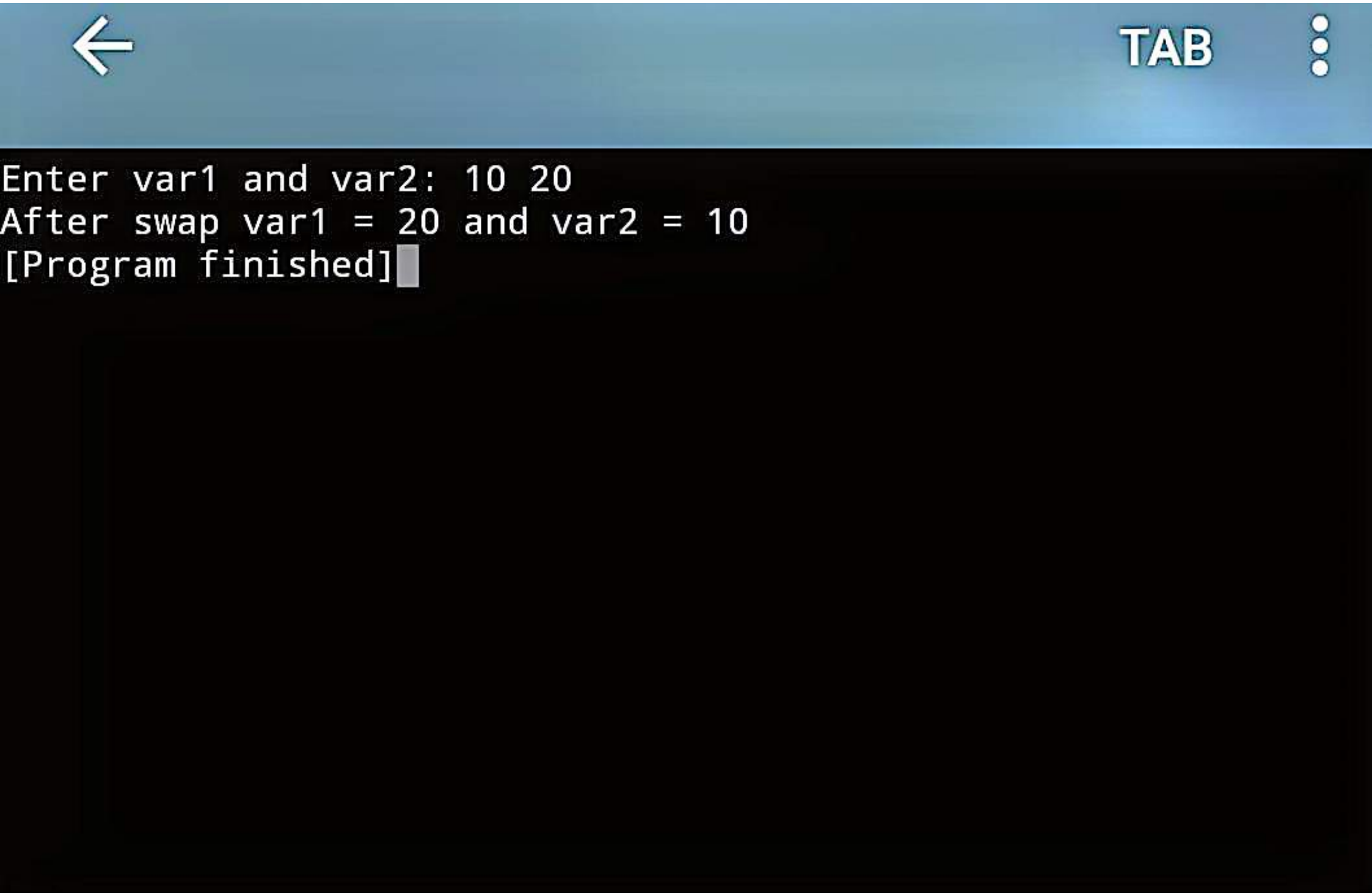


CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int var1, var2, temp;
6      cout << "Enter var1 and var2: ";
7      cin >> var1 >> var2;
8
9      temp = var1;
10     var1 = var2;
11     var2 = temp;
12
13     cout << "After swap var1 = " << var1 << " and
       var2 = " << var2;
14     return 0;
15 }
```

CamScanner


TAB
Enter var1 and var2: 10 20
After swap var1 = 20 and var2 = 10
[Program finished]

CamScanner

Home task 1:

Swap two variables.

Input:

```
# include <iostream>
using namespace std;
int main ( ) {
int var 1 , var 2, var 3 temp;
cout << Enter var 1 and var 2 : " ;
cin >> var 1 >> var 2;
temp = var 1;
var 1 = var 2;
var 2 = temp;
cout << " After swap var 1 = " << var 1 << "
and var 2 = " << var 2 ;
```

Date:

return 0;
}

Output:

Enter Var 1 and Var 2: 10 20

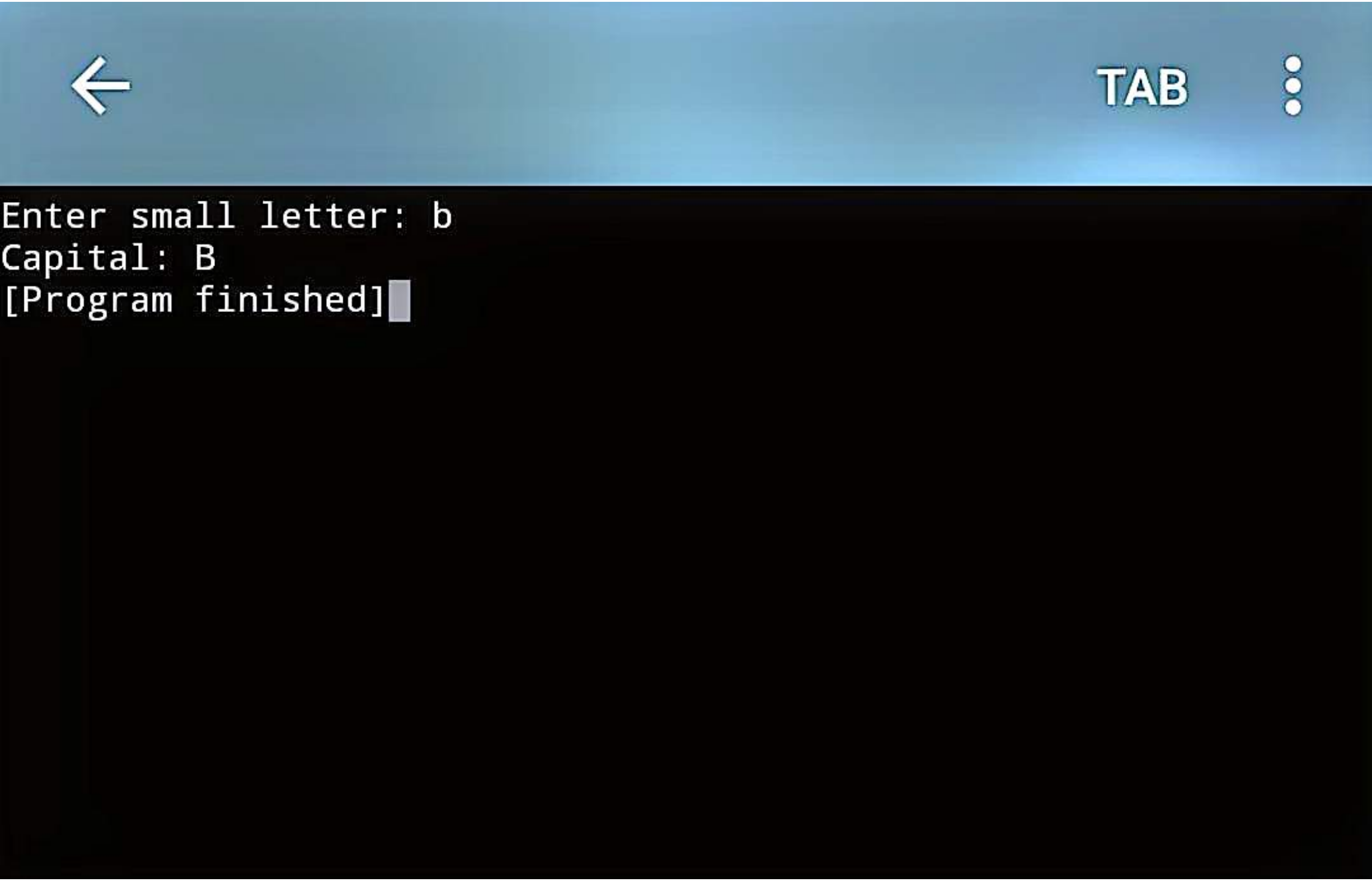
After Swap Var 1 = 20 and Var 2 = 10.

CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      char small;
6      cout << "Enter small letter: ";
7      cin >> small;
8
9      char capital = small - 32; // ASCII trick: a=97,
       A=65, diff 32
10     cout << "Capital: " << capital;
11
12     return 0;
13 }
```

CamScanner



CamScanner

Home task 2

Small to capital. (ASCII)

Input:

```
#include <iostream>
Using namespace std;
int main ( ) {
    char small;
cout << " Enter small letter: ";
        cin >> small;
char capital = small - 32; // ASCII trick:
  a = 97, A = 65, dif 32
  cout << "Capital:" << capital;
        return 0;
}
```

Date:

Date: _____

Output:

Enter small letter : b
Capital : B.

CamScanner

Date: _____

Output:

Enter small letter : b
Capital : B.

CamScanner